

Use Case Specification: Pay Order

1. Use Case Code

UC002

2. Brief Description

This use case describes how the Customer pays for a placed order using a VietQR QR code. It is included by the use case "Place Order". If the customer opts for credit card payment, the use case "Pay Order via PayPal" is triggered instead.

3. Actors

3.1. Customer

The end-user who scans the QR code via a banking application to complete payment, or opts to switch to PayPal.

3.2. VietQR

External payment service provider. AIMS calls VietQR's API to generate the QR code and receives a payment callback with the transaction result.

3.3. emailSender

Internal component responsible for composing and delivering the invoice and transaction confirmation email to the customer upon successful payment.

4. Preconditions

- The use case "Place Order" has been initiated and the customer has confirmed the invoice.
- The invoice — including product list, quantities, prices, VAT, delivery fee, and total amount — has been calculated and temporarily saved by AIMS software.
- AIMS software has valid VietQR API credentials configured.

5. Basic Flow of Events

Step 1. AIMS software calls the VietQR API to generate a QR code. (See Table B – Input data for QR code generation.)

Step 2. VietQR returns the QR code to AIMS software.

Step 3. AIMS software displays the payment screen to the customer. (See Table A – Output data of payment screen.)

Step 4. The customer scans the QR code using their banking application and completes the transfer. (Note: In the sandbox environment, the customer triggers VietQR's test callback API to simulate a successful payment.)

Step 5. VietQR sends a payment callback to AIMS software with the transaction result. (See Table C – Input data from VietQR payment callback.)

Step 6. AIMS software verifies that the callback status is "success" and that the confirmed amount matches the invoice total. AIMS creates an Invoice and a PaymentTransaction record, then displays the payment result screen to the customer and sends a confirmation email.

6. Alternative Flows

Table 1 – Alternative flows of events for UC Pay Order

N o	Location	Condition	Action	Resume Location
1	At Step 1	If the customer chooses to pay via credit card (PayPal) before AIMS calls VietQR	AIMS software inserts the use case "Pay Order via PayPal".	Use case ends
2	At Step 1	If VietQR fails to generate the QR code (e.g., network error, service unavailable)	AIMS software notifies the customer that the payment service is temporarily unavailable. The customer may retry or exit.	Use case ends
3	At Step 3	If the customer wishes to adjust items or delivery information	AIMS software returns the customer to the delivery information / cart screen. The customer updates the order. AIMS recalculates the delivery fee and updates the invoice.	Resumes at Step 8 of UC Place Order
4	At Step 4	If VietQR callback reports payment as failed	AIMS software notifies the customer that the payment was unsuccessful (showPaymentFailed). The customer may: (a) Retry — AIMS re-generates a new QR code and restarts from Step 1; or (b) Cancel — AIMS discards the session and returns to UC Place Order.	Retry: Step 1 Cancel: UC Place Order
5	At any step	If the customer cancels or exits during the payment process	AIMS software calls cancelPayment, discards the session (discardSession), and notifies cancellation (notifyCancellation). No payment is recorded.	Resumes at UC Place Order

7. Input Data

7.1. Input data for QR code generation request (AIMS to VietQR)

Table B – Input data for VietQR QR code generation

N o	Data Field	Description	Mandatory	Valid Condition	Example
1	bank_id	Bank identifier of the receiving account	Yes	Valid VietQR bank code	970422
2	account_no	Receiving bank account number	Yes	Non-empty numeric string	0123456789
3	amount	Total amount to be paid (VND), equal to invoice total	Yes	Positive integer	123456
4	add_info	Transfer description / transaction content	Yes	Non-empty string, max 25 chars	AIMS Order #00123

N o	Data Field	Description	Mand atory	Valid Condition	Example
5	callback_url	URL where VietQR sends the payment result callback	Yes	Valid HTTPS URL	https://aims.example.com/callback

7.2. Input data from VietQR payment callback (VietQR to AIMS)

Table C – Input data from VietQR payment callback

N o	Data Field	Description	Mand atory	Valid Condition	Example
1	transaction_id	Unique transaction identifier assigned by VietQR	Yes	Non-empty string	TXN20231005001
2	transaction_content	Transfer description confirmed by VietQR	Yes	Non-empty string	AIMS Order #00123
3	transaction_datetime	Date and time the transaction was completed	Yes	dd/mm/yyyy hh:mm:ss	05/10/2023 14:30:00
4	amount	Amount received and confirmed by VietQR	Yes	Positive integer, must match invoice total	123456
5	status	Result status of the payment transaction	Yes	"success" or "failed"	success

8. Output Data

8.1. Output data of payment screen (AIMS to Customer)

Table A – Output data of payment screen

N o	Data Field	Description	Display Format	Example
1	QR Code Image	QR code image generated by VietQR for the customer to scan	Image rendered on screen	[QR image]
2	Bank name	Name of the receiving bank	String	MB Bank
3	Account number	Receiving bank account number	Numeric string	0123456789
4	Transfer content	Transfer description corresponding to the order	String	AIMS Order #00123
5	Product title	Title of each media product in the order	String	DVD Mat Biec
6	Quantity	Quantity of each product	Integer, right-aligned	2
7	Unit price	Price per unit (excl. VAT)	Integer with thousands separator, right-aligned	1,234,500
8	Amount	Line total (unit price × quantity, excl. VAT)	Integer with thousands separator, right-aligned	2,469,000
9	Total product price (excl. VAT)	Sum of all line amounts before VAT	Integer with thousands separator, right-aligned	2,469,000
10	Total product price (incl. VAT)	Total product price with 10% VAT applied	Integer with thousands separator, right-aligned	2,715,900

N o	Data Field	Description	Display Format	Example
1 1	Delivery fee	Shipping fee based on weight and location. Not subject to VAT.	Integer with thousands separator, right-aligned	36,360
1 2	Total amount to be paid	Total = total product price (incl. VAT) + delivery fee	Integer with thousands separator, right-aligned	2,752,260

8.2. Output data of payment result screen (AIMS to Customer)

Table A2 – Output data of payment result screen

N o	Data Field	Description	Display Format	Example
1	Customer name	Full name of the customer who placed the order	String	Nguyen Van A
2	Phone number	Customer's contact phone number	Numeric string	0912345678
3	Shipping address	Street address for delivery	String	123 Le Loi, Hoan Kiem
4	Province	Province / city of the delivery address	String	Ha Noi
5	Total amount	Total amount paid by the customer	Integer with thousands separator, right-aligned	2,752,260
6	Transaction ID	Unique transaction identifier from VietQR	String	TXN20231005001
7	Transaction content	Transfer description confirmed by VietQR	String	AIMS Order #00123
8	Transaction datetime	Date and time the transaction was completed	dd/mm/yyyy hh:mm:ss	05/10/2023 14:30:00

9. Postconditions

9.1. Success

- The payment transaction is confirmed as successful.
- AIMS software creates an Invoice record (items, fees) and a PaymentTransaction record (transaction ID, transaction content, transaction datetime).
- The payment result screen is displayed to the customer, showing: customer name, phone number, shipping address, province, total amount paid, transaction ID, transaction content, and transaction datetime. (See Table A2 – Output data of payment result screen.)
- A confirmation email containing the invoice and payment transaction information is sent to the customer.
- The order is placed in the "pending processing" state, awaiting review by the product manager.

9.2. Failure / Cancellation

- If payment fails (VietQR callback status = "failed"), AIMS notifies the customer. The customer may retry (AIMS restarts from Step 1 with a new QR code) or cancel.
- If the customer cancels or exits at any point, AIMS discards the session. No payment record and no order are saved.

- Control returns to UC Place Order at the invoice screen.
- Note: VietQR does not support an automated refund API. If an order that was successfully paid is subsequently cancelled, the product manager is responsible for issuing the refund manually.